LAPORAN PRAKTIKUM INTERNET OF THINGS

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POLITEKNIK NEGERI LAMPUNG

JURUSAN TEKNOLOGI INFORMASI

PROGRAM STUDI TEKNOLOGI REKAYASA INTERNET

2026

7.7 Tugas: Pengamanan Server IoT Industri (MQTT Authentication & Access Control)

C:\Windows\System32\cmd.exe

```
D:\Docker\iot-server>docker exec -it mqtt-broker mosquitto_passwd /mosquitto/config/password.txt admin
Warning: File /mosquitto/config/password.txt has world readable permissions. Future versions will refuse to load this file.
To fix this, use `chmod 0700 /mosquitto/config/password.txt`.
Password:
Reenter password:
Adding password for user admin

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug mqtt-broker
    Learn more at https://docs.docker.com/go/debug-cli/

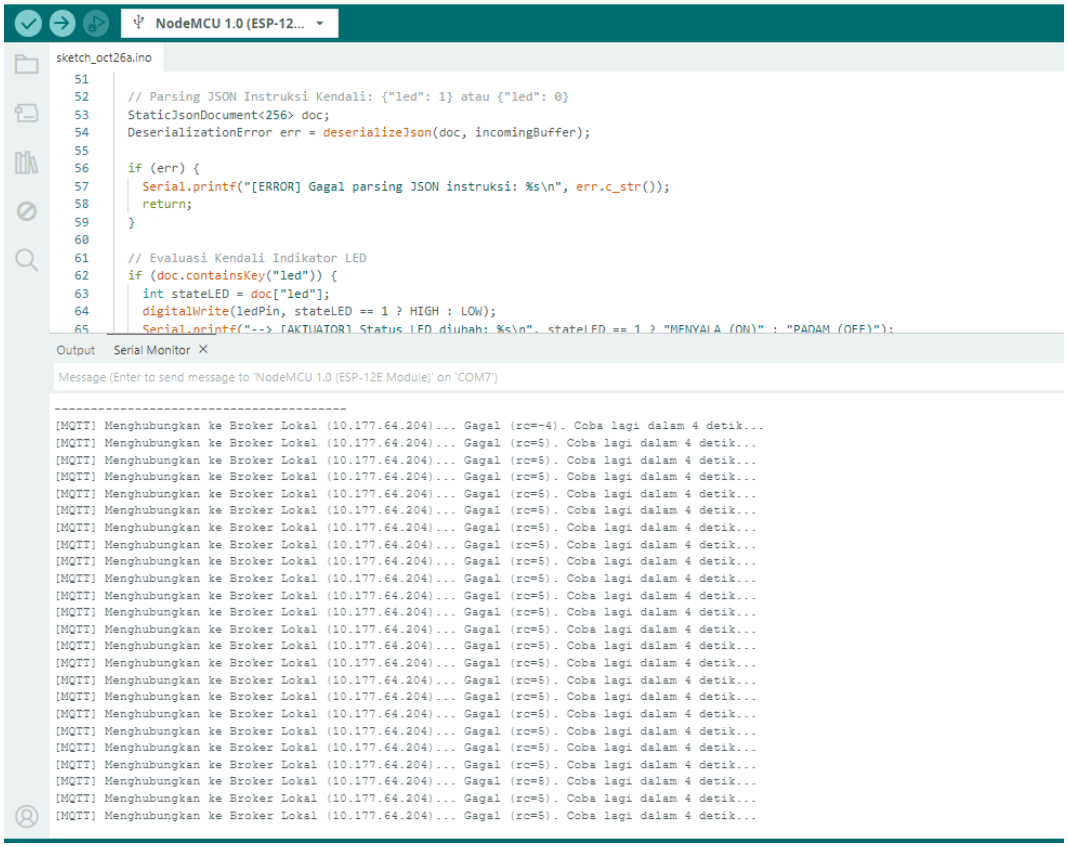
D:\Docker\iot-server>
```

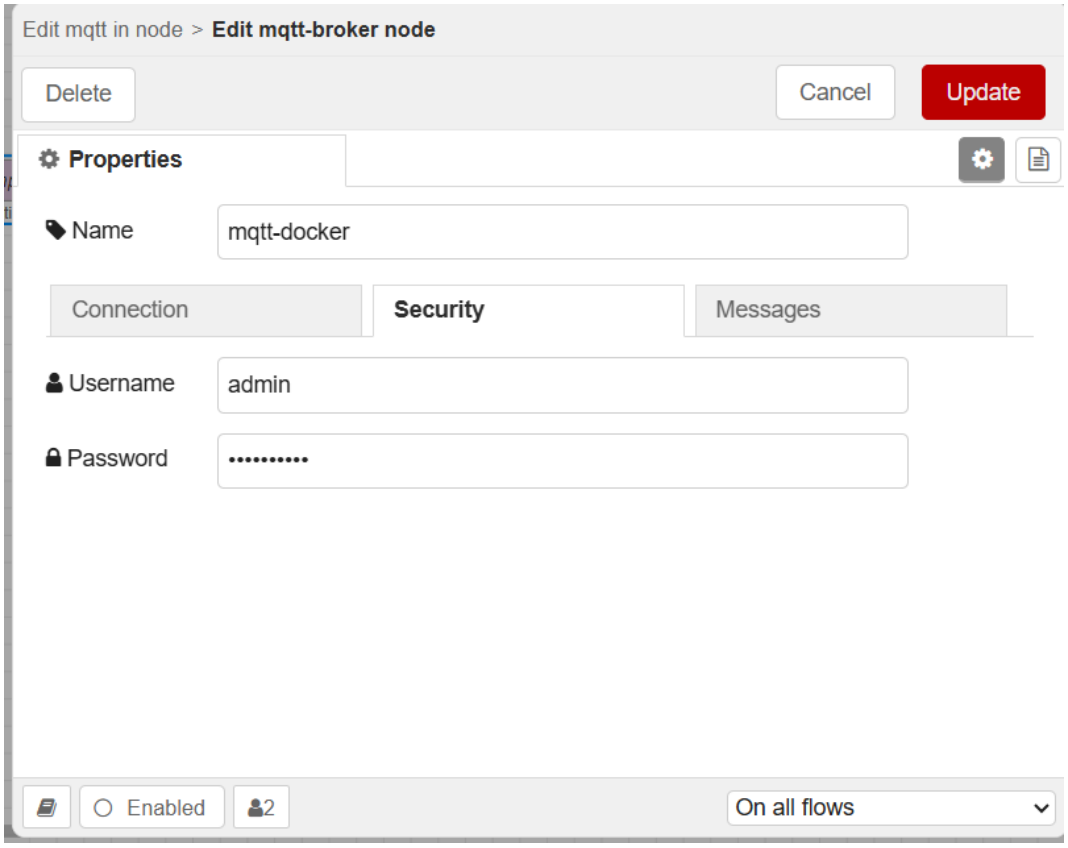
```
D:\Docker\iot-server>docker restart mqtt-broker
mqtt-broker

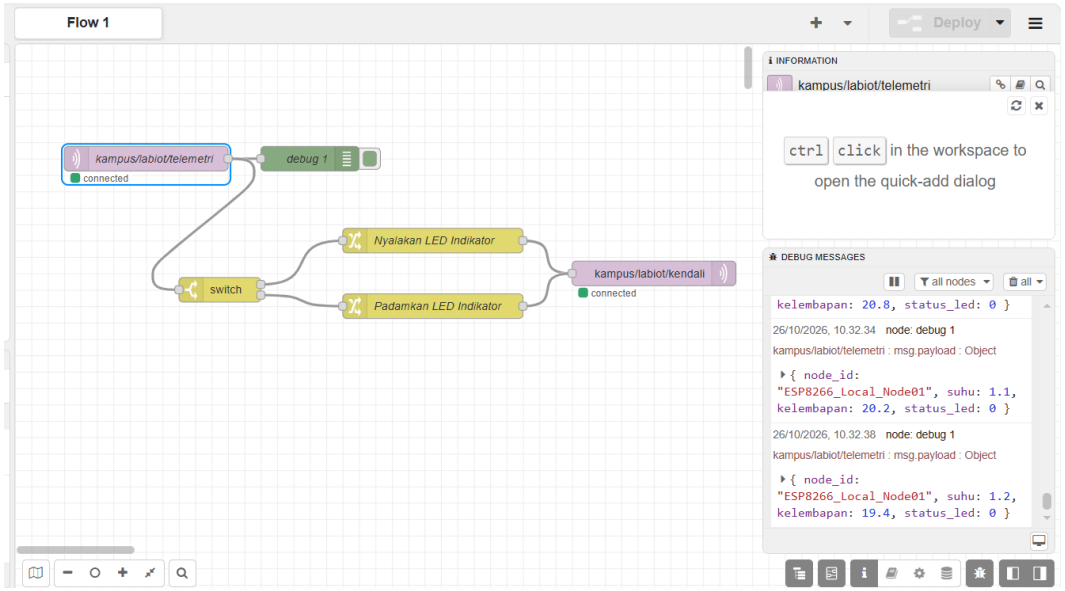
D:\Docker\iot-server>
```

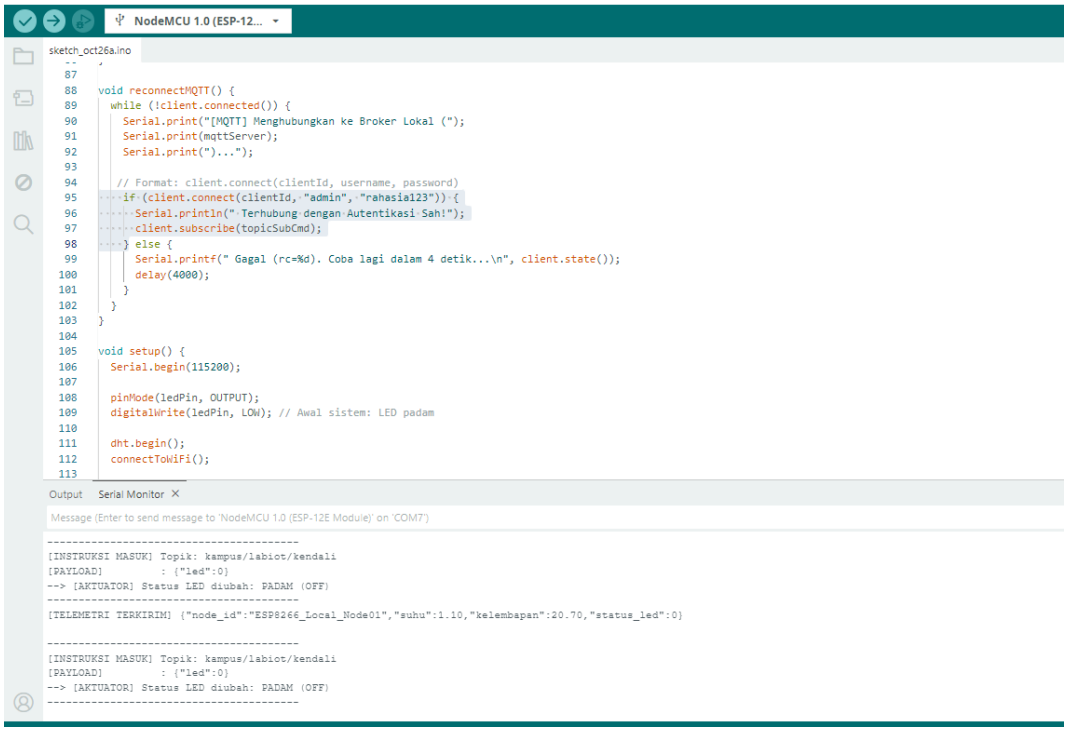
```
# Perizinan Akses Awal (Tanpa Password)
allow_anonymous false
password_file /mosquitto/config/password.txt
```

Name	Date modified	Type	Size
mosquitto.conf	26/10/2026 10:27	CONF File	1 KB
password.txt	26/10/2026 10:25	Text Document	1 KB


NodeMCU 1.0 (ESP-12...
sketch_oct26a.ino
// Parsing JSON Instruksi Kendali: {"led": 1} atau {"led": 0}
StaticJsonDocument<256> doc;
DeserializationError err = deserializeJson(doc, incomingBuffer);
if (err) {
Serial.printf("[ERROR] Gagal parsing JSON instruksi: %s\n", err.c_str());
return;
}
// Evaluasi Kendali Indikator LED
if (doc.containsKey("led")) {
int stateLED = doc["led"];
digitalWrite(ledPin, stateLED == 1 ? HIGH : LOW);
Output
Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM7')
[MQTT] Menghubungkan ke Broker Lokal (10.177.64.204)... Gagal (rc=-4). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (10.177.64.204)... Gagal (rc=5). Coba lagi dalam 4 detik...


Edit mqtt in node > Edit mqtt-broker node
Delete
Cancel
Update
Properties
Name
mqtt-docker
Connection
Security
Messages
Username
admin
Password
Enabled
2
On all flows


Flow 1
Deploy
kampus/labiot/telemetri
connected
debug 1
Nyalakan LED Indikator
switch
Padamkan LED Indikator
kampus/labiot/kendali
connected
INFORMATION
kampus/labiot/telemetri
ctrl click in the workspace to open the quick-add dialog
DEBUG MESSAGES
all nodes
all
kelembapan: 20.8, status_led: 0 }
26/10/2026, 10.32.34 node: debug 1
kampus/labiot/telemetri : msg.payload : Object
{ node_id: "ESP8266_Local_Node01", suhu: 1.1, kelembapan: 20.2, status_led: 0 }
26/10/2026, 10.32.38 node: debug 1
kampus/labiot/telemetri : msg.payload : Object
{ node_id: "ESP8266_Local_Node01", suhu: 1.2, kelembapan: 19.4, status_led: 0 }


NodeMCU 1.0 (ESP-12...
sketch_oct26a.ino
87
88 void reconnectMQTT() {
89 while (!client.connected()) {
90 Serial.print("[MQTT] Menghubungkan ke Broker Lokal (");
91 Serial.print(mqttServer);
92 Serial.print(")...");
93
94 // Format: client.connect(clientId, username, password)
95 if (client.connect(clientId, "admin", "rahasia123")) {
96 Serial.println(" Terhubung dengan Autentikasi Sah!");
97 client.subscribe(topicSubCmd);
98 } else {
99 Serial.printf(" Gagal (rc=%d). Coba lagi dalam 4 detik...\n", client.state());
100 delay(4000);
101 }
102 }
103 }
104
105 void setup() {
106 Serial.begin(115200);
107
108 pinMode(ledPin, OUTPUT);
109 digitalWrite(ledPin, LOW); // Awal sistem: LED padam
110
111 dht.begin();
112 connectToWiFi();
113
Output
Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM7')
[INSTRUKSI MASUK] Topik: kampus/labiot/kendali
[PAYLOAD] : {"led":0}
--> [AKTUATOR] Status LED diubah: PADAM (OFF)
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":1.10,"kelembapan":20.70,"status_led":0}
[INSTRUKSI MASUK] Topik: kampus/labiot/kendali
[PAYLOAD] : {"led":0}
--> [AKTUATOR] Status LED diubah: PADAM (OFF)